AMINE MABROUKI

Electrical & Control Systems Engineer | IoT · Renewable Energy · Industrial Computing

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PROFILE

Electrical and control systems engineer from ENIG (industrial computing track), focused on IoT and renewable energy. Final-year project at STEG: design of a complete monitoring solution for a 10.8 kWp solar PV installation, from the ESP32 to the mobile app, with automatic detection of production losses. Industrial experience at STEG, CPG and Yazaki.

EDUCATION

National Engineering Degree — Electrical and Control Systems Engineering, Industrial Computing track *2023 – 2026*

National Engineering School of Gabès (ENIG)

Preparatory Classes for Engineering Schools — IPEIEM, El Manar, Tunis *2019 – 2023*

Baccalaureate in Technical Sciences — Lycée 9 Avril, Moularès *2019*

PROFESSIONAL EXPERIENCE

Engineering Intern — CPG, Gafsa Phosphate Company, Moularès *Jul – Aug 2026 (2 months)*

- Studied starting methods for the washing plant's induction motors: direct-on-line, star-delta, soft starter and variable frequency drive.
- Calculated current, torque and voltage drop; compared the options and recommended a solution for a 75 kW pump.
- Followed preventive and corrective maintenance work in line with safety procedures (lockout/tagout).

Final-Year Project Intern — STEG, Tunisian Electricity and Gas Company, Metlaoui *Feb – May 2026 (4 months)*

- Sized a 10.8 kWp grid-connected PV installation (18 modules, Solis 12 kW inverter) and ran a PVsyst simulation: month-by-month reference production (performance ratio of 88%).
- Developed **Sphere Solar**, an IoT monitoring chain: ESP32 firmware (C/C++) designed to read the inverter over RS485/Modbus RTU, MQTT transmission, FastAPI/SQLite server (REST API, WebSocket, JWT) and React Native/Expo mobile app.
- Detected gradual losses (soiling) through a daily comparison of measured production with the PVsyst reference; alerts, PDF reports and an LLM-based analysis assistant.
- Validated the chain end to end in simulation (Wokwi), ready for connection to the real inverter; secure broker configuration (TLS, ACL) prepared for deployment.

Shop-Floor Intern — Yazaki, automotive wiring, Gafsa *Jul 2024 (1 month)*

- Introduction to wiring harness manufacturing; took part in quality control and first-level maintenance.

ACADEMIC PROJECTS

- **Water level controller** — Arduino, HC-SR04 ultrasonic sensor, relay and pump: automatic control with high and low thresholds.
- **Analog LED clock** — Arduino Nano, DC motor and driver: minutes displayed by LEDs and clock hands driven by the motor.

TECHNICAL SKILLS

Energy & electrical engineering: solar PV (PVsyst, system sizing), electrical machines and motor starting, PSIM, Proteus
Automation & control: TIA Portal, Ladder, PID control, MATLAB/Simulink, LabVIEW
Embedded & IoT: ESP32, STM32, Arduino, RS485/Modbus RTU, MQTT, FPGA/VHDL, assembly
Software: C/C++, Python (FastAPI), Java, SQL/SQLite, React Native, Android Studio, Git/GitHub

CERTIFICATIONS

- **QHSE certification (Quality, Health, Safety, Environment)** — Centre Beta, 2026
- 5 certificates in programming and algorithms (C++, problem solving) — Programming Advices

LANGUAGES, STUDENT LIFE AND INTERESTS

Languages: Arabic (native), French (good command, B2), English (good command, B2)
Student life: ENIGROBOTS 5.0 (ENIG, 2024) — technical community (preparing the competition model) and logistics.
Interests: reading, sport, chess.